



Name: \_\_\_\_\_ Cohort/Batch: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_

## LANGGRAPH PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A AI Agents

TOTAL: 10 QUESTIONS

**Q1** What is LangGraph and what AI/ML use case is it designed for?

Threat Model

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**Q6** How do you evaluate a model's performance in LangGraph?

Observability

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**Q2** What are the core abstractions in LangGraph?

Architecture

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**Q7** How do you deploy a model trained with LangGraph?

Lifecycle

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**Q3** How does LangGraph handle training or inference?

Configuration

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**Q8** How does LangGraph handle distributed or multi-GPU training?

Compliance

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**Q4** How does LangGraph manage data preprocessing?

Hardening

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**Q9** How do you track experiments and versions in LangGraph?

Testing

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**Q5** How does LangGraph support GPU/TPU acceleration?

SSO / IdP

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**Q10** What are common pitfalls when using LangGraph in production?

Tuning

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ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 LangGraph is a framework or service

Mitigation

LangGraph is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 LangGraph organizes work around abstractions

Primitives

LangGraph organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, LangGraph defines a model

Hardening

For training, LangGraph defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 LangGraph provides data-parallel or

Governance

LangGraph provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 LangGraph provides data loaders, tokenizers, or

Gotchas

LangGraph provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside LangGraph to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 LangGraph detects available accelerators and

Federation

LangGraph detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.